

PySensemakr: Sensitivity Analysis Tools for OLS in Python

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Sensitivity Analysis for Omitted Variable Bias

The most common strategy for drawing causal inferences from observational data is to adjust for observed covariates in a regression model, while making the untestable assumption that there are no unobserved confounders. Since this assumption cannot be verified with the data at hand, sensitivity analysis—which asks how strong unobserved confounders would need to be in order to change the conclusions of a study—plays a central role in determining the credibility of causal claims based on observational data.

Cinelli and Hazlett (2020) proposes a modern suite of sensitivity analysis tools based on a partial R^2 parameterization of omitted variable bias. These tools (i) do not require assumptions on the functional form of the treatment assignment mechanism nor on the distribution of the unobserved confounders, (ii) naturally handle multiple confounders, possibly acting non-linearly, (iii) exploit expert knowledge to bound sensitivity parameters using observed covariates as benchmarks, and (iv) can be easily computed using only standard regression output. The methodology thus lends itself naturally to *routine reporting*—the practice of including sensitivity statistics alongside standard regression tables, much like one would report p -values or confidence intervals.

PySensemakr is a Python package that implements this full suite of sensitivity analysis tools for linear regression models. The package allows researchers to (i) compute sensitivity statistics for routine reporting, such as the partial R^2 of the treatment with the outcome and the robustness value, (ii) derive bounds on the strength of unobserved confounders using observed covariates as benchmarks, and (iii) produce sensitivity contour plots and extreme scenario plots that provide a visual summary of the sensitivity of causal estimates to unobserved confounding. **PySensemakr** is built on top of **statsmodels** (Seabold and Perktold 2010) and requires only standard regression output, without further assumptions on the functional form of the treatment assignment mechanism or on the distribution of the unobserved confounders.

The R version of **sensemakr** (Cinelli, Ferwerda, and Hazlett 2024) has been widely adopted across disciplines—including political science, economics, epidemiology, and education—and has been cited in thousands of empirical studies. A Stata version is also available. While these implementations serve the R and Stata communities well, Python has lacked an equivalent tool, despite being one of the most widely used languages for data analysis across the social sciences, biomedical research, and industry. **PySensemakr** fills this gap, bringing the full **sensemakr** methodology to the Python ecosystem and lowering the barrier for researchers who wish to incorporate formal sensitivity analysis into Python-based regression workflows.

Functionality

PySensemakr takes a fitted **statsmodels** **OLSResults** object as input and computes all relevant sensitivity statistics from the standard regression output. The package is organized into the following modules:

- **main:** The **Sensemakr** class serves as the primary interface. It takes a fitted model, a treatment variable, and optional benchmark covariates, and computes sensitivity statistics and bounds. It provides **summary()** and **plot()** methods for reporting results.
- **sensitivity_statistics:** Functions for computing the partial R^2 of the treatment with the outcome, the robustness value RV_q (the minimum confounding strength required to reduce the estimate by a fraction q), and the robustness value $RV_{q,\alpha}$ (accounting for statistical significance).

- **sensitivity_bounds**: Functions for computing bounds on the bias-adjusted estimates, using observed covariates as benchmarks for the plausible strength of unobserved confounders.
- **sensitivity_plots**: Functions for producing contour plots of bias-adjusted estimates and t -values as functions of the sensitivity parameters $R^2_{D \sim Z | \mathbf{X}}$ and $R^2_{Y \sim Z | \mathbf{X}, D}$, as well as extreme scenario plots.
- **bias_functions**: Core functions for computing the bias, bias-adjusted estimates, standard errors, and t -values.

The following code illustrates a typical workflow using the example dataset of Hazlett (2020):

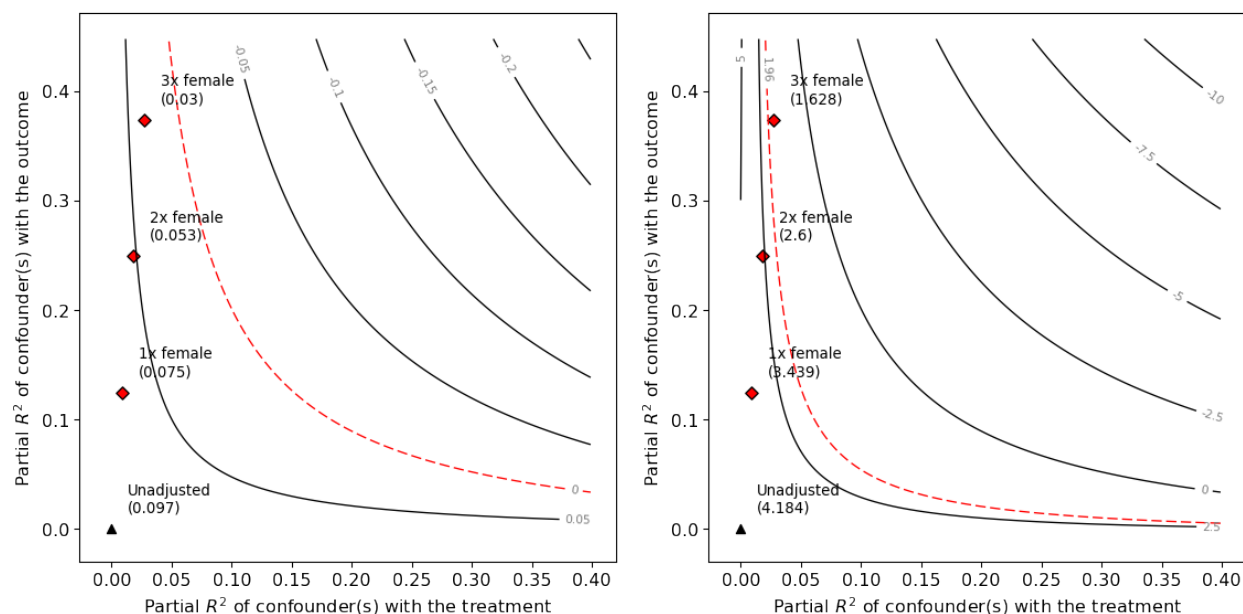
```
import sensemakr as smkr
import statsmodels.formula.api as smf

# Load data and fit regression model
darfur = smkr.load_darfur()
model = smf.ols(formula='peacefactor ~ directlyharmed + age + farmer_dar + '
                'herder_dar + pastvoted + hhsize_darfur + female + village',
                data=darfur).fit()

# Run sensitivity analysis
sense = smkr.Sensemakr(model=model,
                        treatment="directlyharmed",
                        benchmark_covariates=["female"],
                        kd=[1, 2, 3])

# Print and plot results
sense.summary()
sense.plot()
```

?? illustrates the contour plot output, in which the estimate and the t -value of the treatment effect are displayed as functions of the hypothetical confounding strength. Figure 1 shows the corresponding extreme scenario plot.



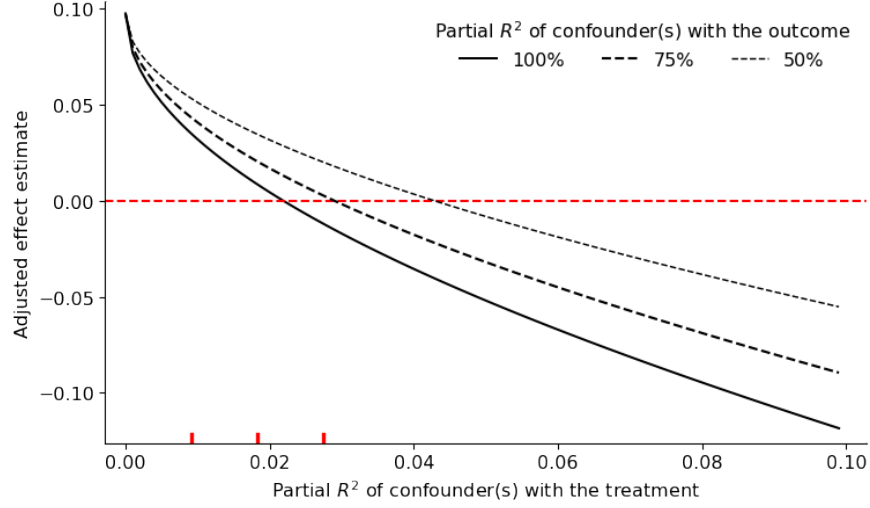


Figure 1: Extreme scenario plot for the Darfur data, showing the bias-adjusted estimate under the worst-case scenario for confounders of varying strength.

Acknowledgements

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References

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